

GabayGamot

AI-Assisted Medicine Monitoring, Dispensing, and Referral System to Reduce Medicine Waste and Shortages

1. Introduction

1.1 System Overview

GabayGamot is a web-based medicine inventory, dispensing, monitoring, and referral system designed to help barangay health centers manage medicine stocks efficiently and reduce medicine waste.

The system uses camera-based scanning with Optical Character Recognition (OCR) to automatically extract important medicine details such as medicine name, expiry date, and category. After scanning, the user reviews the extracted information and manually inputs the medicine quantity before submission to ensure accuracy.

The system also provides medicine availability monitoring between barangays. If a medicine is unavailable in one barangay, the system searches nearby barangays with available stock and automatically generates a referral form.

GabayGamot also includes a dispensing log feature that records medicines given to patients. This helps improve accountability, monitor medicine usage, and automatically deduct inventory after dispensing.

All validated records are stored in a centralized database. The system integrates Gemini API to provide AI-assisted recommendations for redistribution, shortage prevention, procurement planning, and medicine demand forecasting.

The Admin manages users, monitors inventory, views analytics, and generates reports to support healthcare decision-making.

2. Problem Statement and Proposed Solution

2.1 Problem Statement

Many barangay health centers experience difficulties in managing medicine inventory because of manual recording and disconnected monitoring systems.

These issues often result in inaccurate stock monitoring, medicine shortages, delayed referrals, expired medicines, and poor medicine accountability.

According to COA reports, many medicines expired or remained unused because of poor inventory systems, weak procurement planning, and inefficient monitoring.

Some barangays experience medicine shortages while nearby barangays have excess stocks. Patients also experience delays because health centers cannot easily monitor medicine availability from nearby barangays.

Another challenge is the lack of proper monitoring of medicines dispensed to patients. Without dispensing records, health workers may have difficulty tracking medicine usage and accountability.

2.2 Proposed Solution (GabayGamot)

GabayGamot provides a centralized solution for medicine inventory monitoring, dispensing, patient referrals, and procurement planning.

The system uses OCR-based scanning to automatically extract medicine information such as medicine name, expiry date, and category. Users review the extracted information and manually input medicine quantity before submission.

The system monitors medicine availability, generates expiry alerts, records dispensing logs, automatically deducts inventory after dispensing, and searches nearby barangays with available medicines for referral generation.

Gemini API provides AI-assisted recommendations for redistribution, shortage prevention, procurement planning, and medicine demand forecasting.

The admin dashboard supports monitoring, analytics, and report generation for improved healthcare coordination and decision-making.

3. Objectives

The system aims to improve medicine inventory monitoring, reduce expired medicines, improve barangay coordination, automate referrals, and support smarter procurement planning.

It also aims to reduce manual encoding, improve medicine accountability through dispensing logs, and provide AI-assisted recommendations for better healthcare decision-making.

4. System Scope

GabayGamot includes account authentication, medicine scanning, inventory management, expiry alerts, medicine dispensing, referral generation, AI-assisted recommendations, analytics dashboards, and report generation.

The system is intended for barangay health centers and local health offices.

5. System Users

5.1 Admin

The Admin manages the overall operations of the system.

Responsibilities include:

- managing users
- monitoring inventory
- viewing analytics
- monitoring referrals
- monitoring dispensing logs
- generating reports
- monitoring AI recommendations

5.2 Barangay Health Worker

The Barangay Health Worker performs daily medicine inventory and dispensing operations.

Responsibilities include:

- scanning medicines
- updating inventory
- dispensing medicines
- recording dispensing logs
- generating referrals
- viewing AI recommendations

6. System Features

GabayGamot provides OCR-based medicine scanning, inventory management, dispensing logs, automatic stock deduction, nearby medicine search, referral generation, AI-assisted recommendations, dashboards, analytics, and report generation.

These features help improve medicine tracking, monitoring, accountability, and coordination between barangays.

7. System Workflow

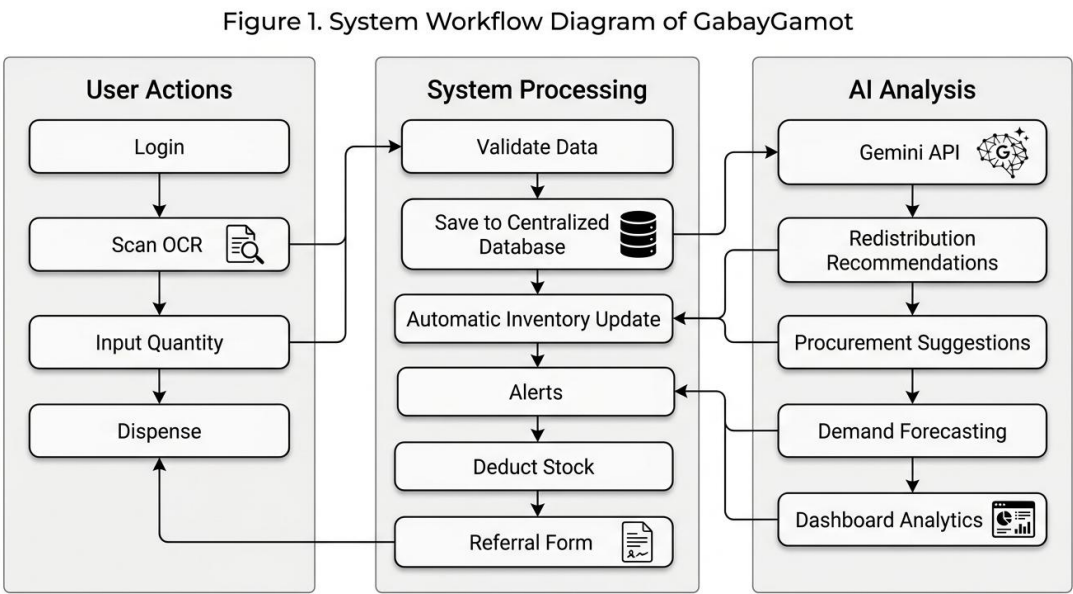
The system follows a structured process starting from admin registration and barangay setup.

Users scan medicines, review extracted data, input quantity, and store validated records in the centralized database.

The system automatically updates inventory, generates expiry alerts, records dispensing logs, deducts stock after dispensing, and generates referrals when medicines are unavailable.

Gemini API analyzes medicine usage and inventory trends to generate procurement and redistribution recommendations.

The admin monitors inventory, referrals, dispensing records, analytics, and reports through the dashboard.



8. System Architecture

The system uses React JS (Vite) for the frontend, Node.js with Express.js for the backend, and Firebase Firestore for database management.

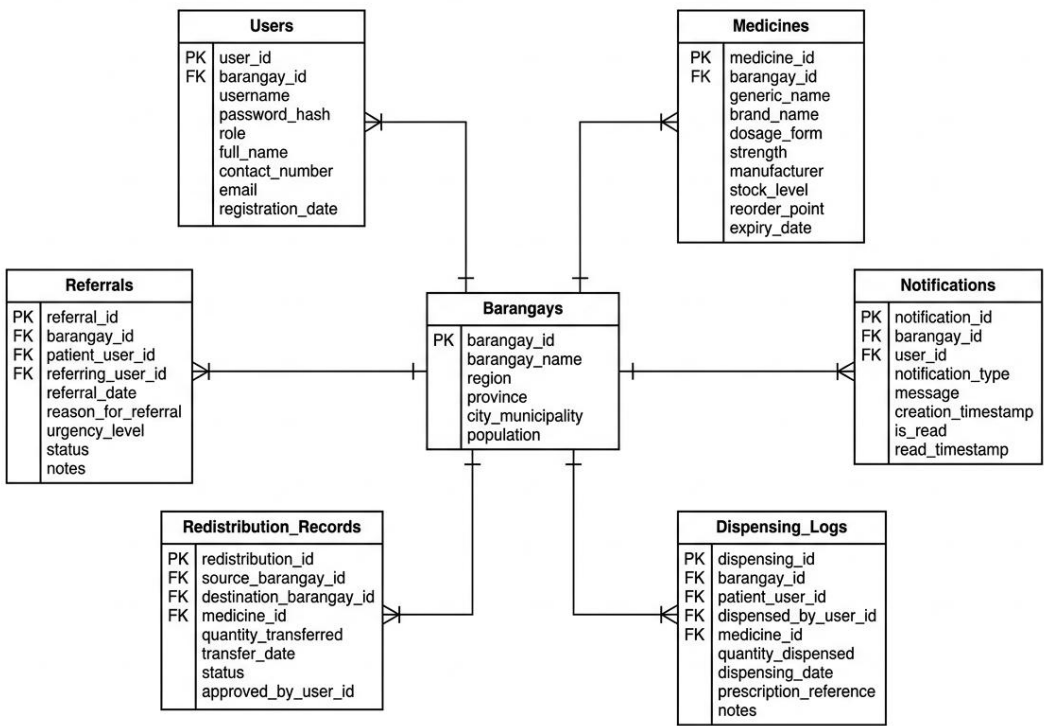
Additional technologies include Firebase Authentication, Tailwind CSS, Shadcn UI, Tesseract.js for OCR scanning, Mapbox API for location services, and Gemini API for AI integration.

9. Database Design

The database is composed of entities including Users, Barangays, Medicines, Referrals, Notifications, Redistribution Records, and Dispensing Logs.

These entities store information related to medicine inventory, dispensing activities, referrals, notifications, and AI-generated recommendations.

Figure 2. Entity Relationship Diagram (Crow’s Foot Notation)
GabayGamot - AI-Assisted Medicine Monitoring, Dispensing, and Referral System



10. Relationships

The system follows structured relationships between entities.

One Barangay can have many Users, Medicines, Referrals, Notifications, and Dispensing Logs.

Each User belongs to one Barangay, while Medicines, Referrals, and Dispensing Logs are associated with a specific Barangay.

11. Data Flow Summary

GabayGamot processes information through input, processing, and output stages.

Inputs include medicine images, medicine quantities, dispensing records, referral requests, and login credentials.

The system processes OCR extraction, validation, inventory monitoring, dispensing, referral generation, and AI recommendation analysis.

Outputs include updated inventory records, dispensing logs, alerts, referrals, procurement recommendations, and reports.

Figure 3. Data Flow Diagram Level 0

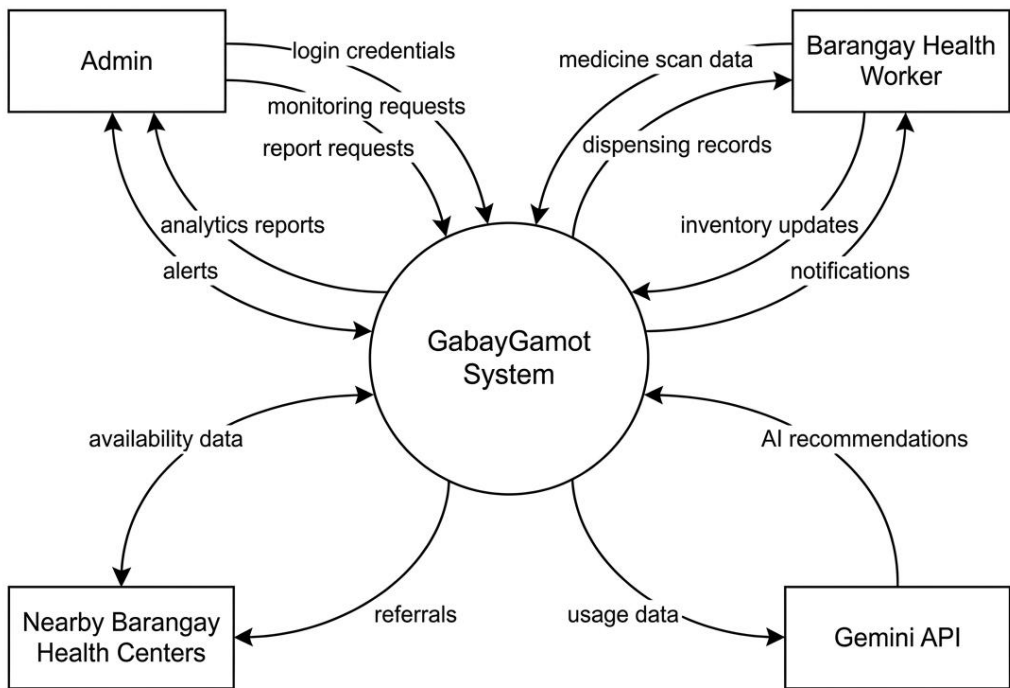
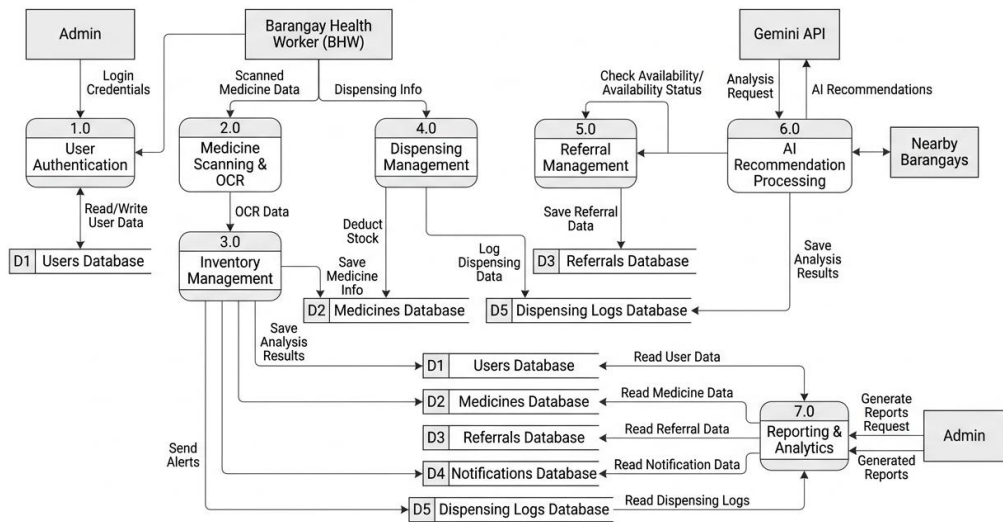


Figure 4. Data Flow Diagram Level 1

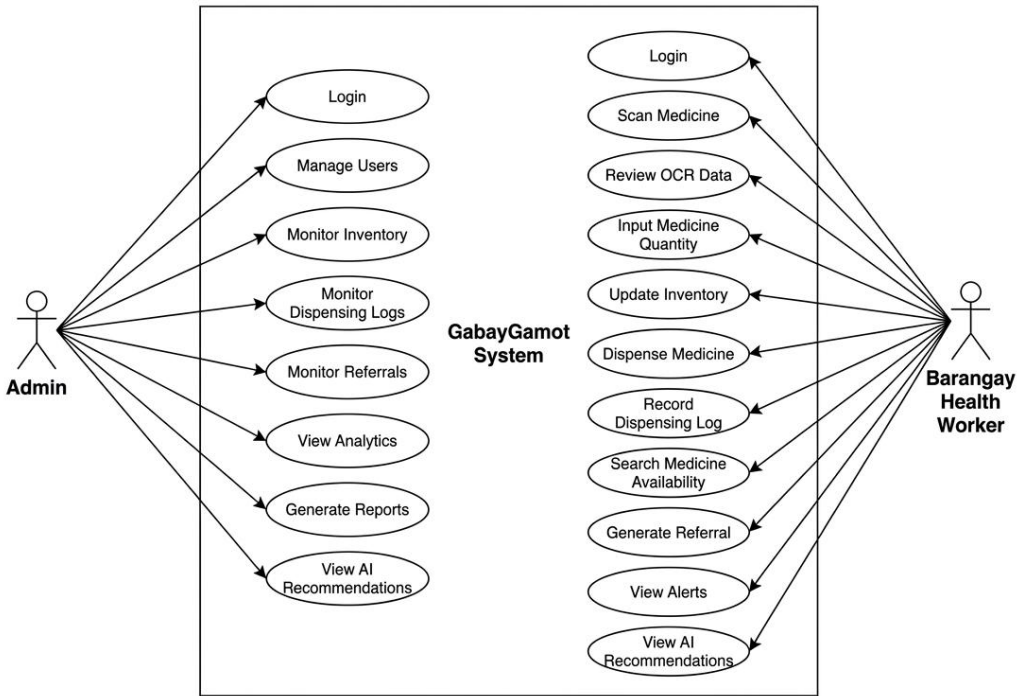


12. Use Case Diagram

The Use Case Diagram illustrates the interaction between the Admin and Barangay Health Worker with the system.

The diagram includes major system functions such as login, inventory management, medicine dispensing, referral generation, AI recommendations, and report generation.

Figure 5. Use Case Diagram of GabayGamot



13. Possible System Pages

Admin Pages

- Login Page
- Dashboard
- User Management
- Inventory Monitoring
- Reports and Analytics

Barangay Health Worker Pages

- Dashboard
- Scan Medicine Page
- Dispensing Page
- Referral Page
- Reports Page

14. Dispensing Log Feature

The Dispensing Log feature records medicines given to patients.

This feature helps improve medicine accountability, track medicine usage, monitor dispensing history, reduce inventory discrepancies, and support patient safety.

The dispensing log records the patient name, medicine name, quantity dispensed, date dispensed, health worker assigned, and remarks.

When medicines are dispensed, the system automatically deducts the quantity from the inventory.

15. System Constraints

The system has some limitations.

- OCR accuracy depends on image quality
- medicine quantity must still be manually entered
- internet connection is required
- Gemini API requires internet access
- location accuracy depends on maps integration
- the system is not a full electronic medical record system

16. System Type

GabayGamot is a web-based system and does not use IoT devices.

17. Future Improvements

Future versions of the system may include disease trend analytics, barcode scanning, mobile application support, predictive analytics, and advanced forecasting.

18. Summary

GabayGamot improves medicine inventory management by combining OCR, centralized monitoring, dispensing logs, AI-assisted recommendations, and barangay coordination.

The system helps reduce expired medicines, improve medicine accessibility, automate referrals, improve procurement planning, and support smarter healthcare decision-making.

By improving visibility and coordination between barangays, GabayGamot helps ensure that medicines are properly utilized instead of expiring unused while nearby communities experience shortages.